

Econometrics STATA Lab: A Mini WASDE–Bloomberg Project

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Week 2

Today: from a research question to a merged dataset and one regression.

Start with a research question

Question

Do USDA WASDE production revisions show up in Bloomberg futures-price surprises on the release date?

- WASDE tells us whether the reported production outlook changed.
- Bloomberg futures prices tell us how markets moved around the release.
- A simple supply story predicts: a positive production revision should be associated with a lower price surprise.

Question first. Code second.

The two files: two views of the same event

WASDE revisions

One row per **release** $\times$ **crop**.

- release_date
- commodity
- production_revision

Bloomberg futures

One row per **release** $\times$ **crop**.

- release_date
- commodity
- price_surprise

Research simplification

price_surprise is a precomputed Bloomberg futures PC1 shock. PCA is part of the research pipeline, not today's lesson.

First: explore the WASDE file

```
use wasde_revisions_week2.dta, clear
describe
list release_date commodity in 1/10
summarize production_revision
tabulate commodity
isid release_date commodity
```

Ask students

What is one observation? What is the unit of analysis?

Date	Crop	Revision
2010-07-09	Corn	-0.83
2010-07-09	Oats	-0.72
2010-07-09	Rice	2.04
2010-07-09	Soybeans	1.00

Then: explore the Bloomberg file

```
use bbg_futures_week2.dta, clear
describe
list commodity price_surprise in 1/10
summarize price_surprise
tabulate commodity
isid release_date commodity
```

Ask students

Does this file have the same observation unit and key?

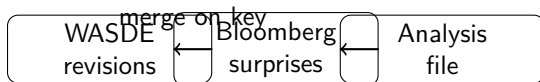
Date	Crop	Surprise
2010-07-09	Corn	-0.30
2010-07-09	Oats	2.91
2010-07-09	Rice	-0.15
2010-07-09	Soybeans	0.85

Why merge these files?

The key is the common question: which crop, on which release?

`release_date + commodity`

- WASDE contributes `production_revision`.
- Bloomberg contributes `price_surprise`.
- After the merge, one row contains both variables needed for the regression.



A join example before the research data

Master	
id	name
1	Ana
2	Ben
3	Chen

Using	
id	score
1	80
2	92
4	71

After merge		
id	name	score
1	Ana	80
2	Ben	92
3	Chen	.
4	.	71

Adapted from UCLA OARC, Combining data in Stata. The point is to see matches and non-matches before memorizing syntax.

Merge type 1: 1:1

Meaning

The key is unique in the master and unique in the using file: one row matches at most one row.

```
use master.dta, clear
isid id
merge 1:1 id using using.dta
tabulate _merge
```

Master	Using	Result
one row for id=1	one row for id=1	one matched row
one row for id=2	no id=2	master-only row
no id=4	one row for id=4	using-only row

`isid` checks the claim behind the merge type.

Merge type 2: 1:m

Meaning

One master row matches many using rows. Think: one release event, many crop observations.

```
use wasde_revisions_week2.dta, clear
keep event_id release_date
bysort event_id: keep if _n == 1
tempfile releases
save 'releases'
merge 1:m event_id using bbg_futures_week2.dta
tabulate _merge
```

Event file	Futures file	Result
2010-07 event	Corn, Oats, Rice, ...	five rows
2010-08 event	Corn, Oats, Rice, ...	five rows

The "m" side is allowed to repeat the key.

Merge type 3: m:1

Meaning

Many master rows match one using row. It is the same relationship viewed from the other direction.

```
use bbg_futures_week2.dta, clear
* event_id repeats here by design: this is the many side
merge m:1 event_id using 'releases'
tabulate _merge
```

Futures file	Event file	Result
five rows for 2010-07	one 2010-07 row	five matched rows
five rows for 2010-08	one 2010-08 row	five matched rows

Choose the merge type by asking: how many rows can each key have on each side?

The important variable: `_merge`

Value	Meaning	What to do in class
1	observation only in master	inspect it
2	observation only in using	inspect it
3	observation in both files	candidate for analysis

Student exercise

Run `tabulate _merge`. Then list the rows with `_merge == 1` and `_merge == 2`. Explain why they are unmatched.

`keep if _merge == 3` is a choice about the sample, not a definition of 1:1, 1:m, or m:1.

Now merge the research files

```
use wasde_revisions_week2.dta, clear
isid release_date commodity

merge 1:1 release_date commodity using bbg_futures_week2.dta

tabulate _merge
list release_date commodity if _merge == 1
list release_date commodity if _merge == 2
```

Expected classroom result

60 matched observations, one master-only observation, and one using-only observation. We inspect first; only then do we keep matched rows.

One regression: test the simple supply story

Specification

$$\text{price_surprise}_{ec} = \alpha + \beta \text{production_revision}_{ec} + \varepsilon_{ec}$$

```
keep if _merge == 3  
drop _merge  
regress price_surprise production_revision
```

- A negative $\hat{\beta}$ is consistent with 'more expected supply, lower price.'
- The coefficient is an association in this simple classroom exercise, not automatically a causal effect.
- Ask: what is the unit of β ? What variation identifies it?

The complete reproducible workflow

- 1 Start with a question.
- 2 Open and describe each file.
- 3 Identify the observation unit and key.
- 4 Merge and audit `_merge`.
- 5 Keep the intended sample.
- 6 Run one simple regression.
- 7 Export a table and interpret the sign, size, and limitations.

User-written command: run once

```
ssc install estout  
esttab main using results.tex, replace tex se
```

Suggested 30–40 minute class plan

Minutes	Activity
0–5	Research question and prediction
5–12	Explore WASDE and Bloomberg files
12–25	Join example; teach 1:1, 1:m, m:1; inspect <code>_merge</code>
25–34	Merge the research files and run the regression
34–40	Interpret the coefficient and open the exported PDF table

Take-home message

Real research is often: define the question, understand the rows, combine sources carefully, and only then estimate.